What we have described so far would not be considered a true actor-critic method by some, because we're only using the value function as a baseline and not using it to "bootstrap" by making a prediction about a future state based on the current state. You will see how bootstrapping comes into play soon.

The policy network has a sensitive loss function that depends on the rewards collected at the end of the episode. If we naively tried to make online updates with the wrong type of environment, we might never learn anything because the rewards might be too sparse.

In Gridworld, which we introduced in chapter 3, the reward is –1 on every move except for the end of the episode. The vanilla policy gradient method wouldn't know what action to reinforce, since most actions result in the same reward of –1. In contrast, a Q-network can learn decent Q values even when the rewards are sparse, because it *bootstraps*. When we say an algorithm bootstraps, we mean it can make a prediction from a prediction.

If we ask you what the temperature will be like two days from now, you might first predict what the temperature will be tomorrow, and then base your 2-day prediction on that (figure 5.3). You're bootstrapping. If your first prediction is bad, your second may be even worse, so bootstrapping introduces a source of *bias*. Bias is a systematic deviation from the true value of something, in this case from the true Q values. On the other hand, making predictions from predictions introduces a kind of self-consistency that results in lower *variance*. Variance is exactly what it sounds like: a lack of precision in the predictions, which means predictions that vary a lot. In the temperature example, if we make our day 2 temperature prediction based on our day 1 prediction, it will likely not be too far from our day 1 prediction.

Figure 5.3. Read from left to right, raw data is fed into a predict temperature model that predicts the next day's temperature. That prediction is then used in another prediction model that predicts day 2's temperature. We can keep doing this, but initial errors will compound and our predictions will become inaccurate for distant predictions.